

WILL VARNER

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EDUCATION

University of Georgia

Bachelor of Science in Computer Science; GPA: 3.8

Athens, GA

Expected May 2027

- **Honors & Leadership:** UGA Presidential Scholar, Zell Miller Scholar, Order of Omega (Top 5%), Roblox Campus Ambassador

EXPERIENCE

Roblox

Software Engineering Intern

San Mateo, CA

May 2026 – August 2026

- Architected and deployed Scenario Intelligence, extracting declarative service contracts from ASTs, telemetry, and PR diffs; scaled across 7 core engineering orgs to merge 500+ reviewed scenarios and 1,000+ assertions into production.
- Co-engineered the enterprise agentic test generation pipeline, integrating behavioral contract ingestion into the agent loop to boost critical behavior test coverage by +36.4% across distributed microservices.
- Built differential semantic evaluation tooling and AGENTS.md governance, deploying PR regression gates that detect lost behaviors, eliminate LLM spec padding, and enforce atomic contract sync.
- Automated drift and vulnerability detection across microservices via self-serve adoption, exposing authorization bypasses on live write paths and remediating 2,000+ lines of defective production code.

Mobility Systems Control Laboratory

Undergraduate AV Research Lead

Athens, GA

January 2025 – Present

- Architected a closed-loop red-teaming pipeline coupling a 20Hz C++ telemetry daemon with CARLA 0.9.15 and SUMO co-simulation via POSIX shared-memory IPC.
- Integrated NVIDIA Alpamayo 2 Super (34B VLA), ingesting multi-camera streams to record trajectory traces and natural-language reasoning tokens at 50ms deterministic tick rates.
- Engineered an adversarial scenario mutator in Python that dynamically perturbs pedestrian crossing trajectories and occlusion angles conditioned on real-time ego vehicle kinematics.
- Evaluated 100+ automated closed-loop trials, benchmarking Failure Yield Rate (FYR) and minimum Time-to-Collision (TTCmin) against static OpenSCENARIO baselines.
- Quantified Reasoning-Action Divergence to detect perception-action decoupling where the model articulated hazards in language tokens but executed collision trajectories.

Leidos

Software Engineering Intern

Atlanta, GA

May 2025 – January 2026

- Migrated CDASim co-simulation infrastructure from ROS 1/Python to ROS 2 Humble (CARLA 0.10.0), optimizing data pipelines to increase simulation execution throughput.
- Ported 15+ backend nodes from CARMA-CARLA Bridge to ROS 2 (`rc1py`), eliminating legacy bridge middleware to achieve direct, sub-5ms communication with the CARMA Platform.
- Engineered `carla_ros2_bridge` in C++ and Python, authoring real-time sensor and vehicle interfaces for CARLA 0.10.0, reviewed for inclusion by the official CARLA open-source team.
- Prototyped a unified XML-RPC protocol replacing TraCI for actor and sensor synchronization, reducing serialization overhead by 50% across interaction pipelines.
- Validated real-time dynamic topic throughput (~20Hz) and message fidelity across distributed simulation nodes via automated CI/CD regression test suites.

PUBLICATIONS

- **W. Varner**, et al., “Closed-Loop Red-Teaming of Vision-Language-Action Models: Dynamic Scenario Mutation and Safety Verification in Autonomous Driving,” *targeted for IEEE Intelligent Vehicles Symposium (IEEE IV)*, 2027.

TECHNICAL SKILLS

- **Languages:** Python, C/C++ (C++17/20), Go, TypeScript, Java, Rust, SQL, Lua
- **Systems & Middleware:** POSIX Shared Memory IPC, ROS 2 (`rc1py`), LibCarla API, gRPC, PyArrow
- **Infrastructure & Tools:** Linux/Unix, Docker, Git, CI/CD (GitHub Actions), AWS, GCP, CARLA, SUMO, TraCI
- **Core Domains:** Agentic Test Generation, Autonomous Vehicles, Closed-Loop Verification, OpenSCENARIO